

Lab Setup, Validation & Initial Attack

Surface Baseline

Project	: NovaShyld Task 1
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Assessment Type	: Vulnerability Assessment & Penetration Testing (VAPT) Lab Setup and Baseline Assessment
Attacker Machine	: Kali Linux 2026.x
Target Machine	: Metasploitable 2
Virtualization Platform	: Oracle VirtualBox

1. Objective

The objective of this task was to establish a secure penetration testing laboratory using virtual machines, validate the functionality of essential security tools, configure an isolated network, and perform an initial attack surface assessment of the target system. The baseline information collected during this phase will be used for future vulnerability assessments and penetration testing activities.

Module 1: Advanced Lab Setup

Environment Configuration

Two virtual machines were created within Oracle VirtualBox.

Kali Linux (Attacker)

Configuration	Value
Operating System	: - Kali Linux
RAM	: - 4 GB
CPU	: - 2 Cores
Role	: - Attacker

Metasploitable 2 (Target)

Configuration	Value
Operating System	:- Metasploitable 2
RAM	:- 2 GB
CPU	:- 1 Core
Role	:- Vulnerable Target

Purpose of an Isolated Environment

The lab environment was configured as an isolated virtual network to ensure that penetration testing activities remain confined within the laboratory. This approach prevents unintended interactions with external systems and allows security testing to be conducted safely without affecting production or public networks.

Module 2:Network Architecture & Segmentation

Network Configuration

The virtual machines were connected using a Host-Only/NAT Network within VirtualBox.

Network Details

Device	IP Address
Kali Linux	10.0.2.x
Metasploitable 2	10.0.2.14

Connectivity Verification

Connectivity between the attacker and target systems was verified using:

- Ping
- ARP Table Verification

Successful responses confirmed that both systems were reachable within the isolated network.

Attacker:

- Kali Linux (10.0.2.x)

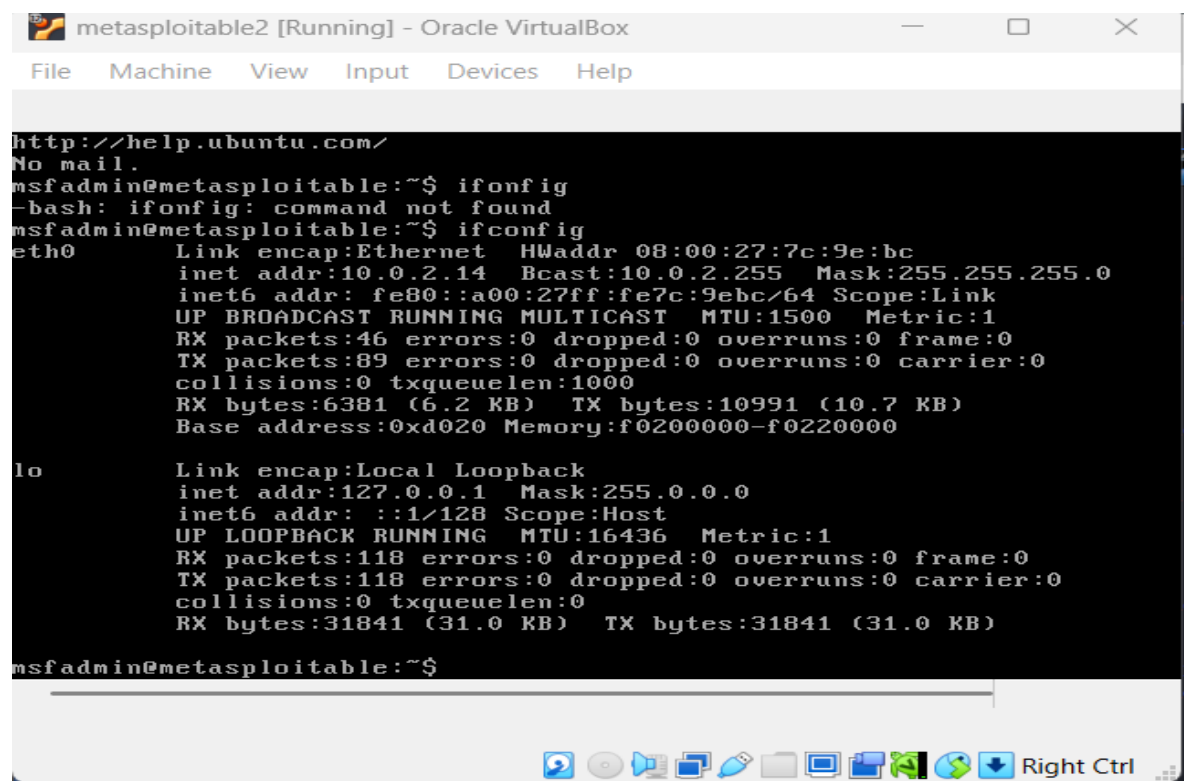
Target:

- Metasploitable 2 (10.0.2.14)

Module 3: Toolchain Validation

The required penetration testing tools were validated before beginning the assessment.

Tool	Purpose	Status
Nmap	Network Discovery & Port Scanning	Verified
Metasploit Framework	Exploitation Framework	Verified
Wireshark	Packet Capture & Analysis	Verified



```
metasploitable2 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
-bash: ifconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:7c:9e:bc
          inet addr:10.0.2.14  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe7c:9ebc/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:46 errors:0 dropped:0 overruns:0 frame:0
          TX packets:89 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:6381 (6.2 KB)  TX bytes:10991 (10.7 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:118 errors:0 dropped:0 overruns:0 frame:0
          TX packets:118 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:31841 (31.0 KB)  TX bytes:31841 (31.0 KB)

msfadmin@metasploitable:~$
```

ARP scanning to discover live hosts in the subnet:-
arp-scan 10.0.2.0/24

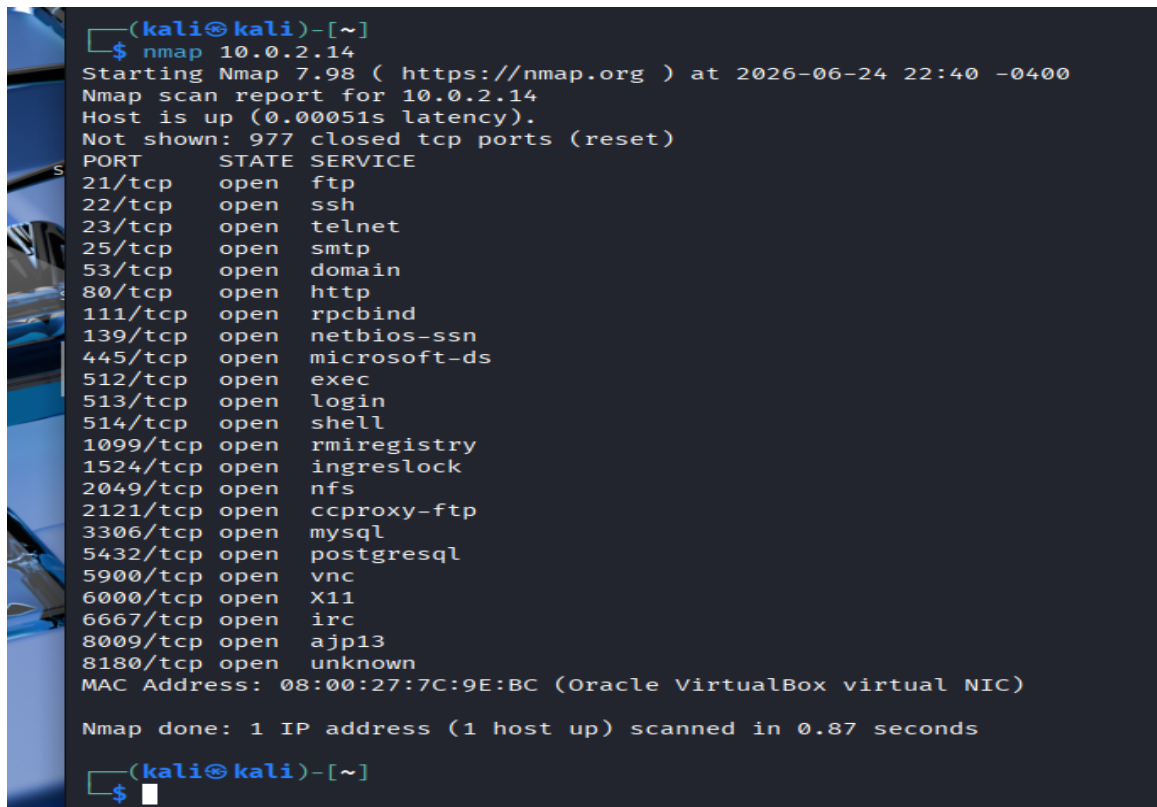
Nmap

Purpose:

- Network discovery
- Port scanning
- Service and OS detection

TCP Port Scan

The scan identified the following open TCP ports.

A terminal window from a Kali Linux machine showing the execution of an Nmap scan on the IP address 10.0.2.14. The output displays the Nmap version (7.98), the scan time (2026-06-24 22:40 -0400), and a list of 21 open TCP ports with their corresponding services. The scan completed in 0.87 seconds.

```
(kali㉿kali)-[~]  
$ nmap 10.0.2.14  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-24 22:40 -0400  
Nmap scan report for 10.0.2.14  
Host is up (0.00051s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:7C:9E:BC (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.87 seconds  
  
(kali㉿kali)-[~]  
$
```

UDP Scan

```
(kali㉿kali)-[~]
$ nmap -sU 10.0.2.14
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-24 23:01 -0400
Nmap scan report for 10.0.2.14
Host is up (0.0021s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open       domain
68/udp    open|filtered dhcp
69/udp    open|filtered tftp
111/udp   open       rpcbind
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open       nfs
MAC Address: 08:00:27:7C:9E:BC (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1506.58 seconds
```

Service Version Detection

```
(kali㉿kali)-[~]
$ nmap -sV 10.0.2.14
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-24 23:01 -0400
Nmap scan report for 10.0.2.14
Host is up (0.00079s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:7C:9E:BC (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.35 seconds
```

Metasploit

Metasploit Framework launched successfully.

Purpose:

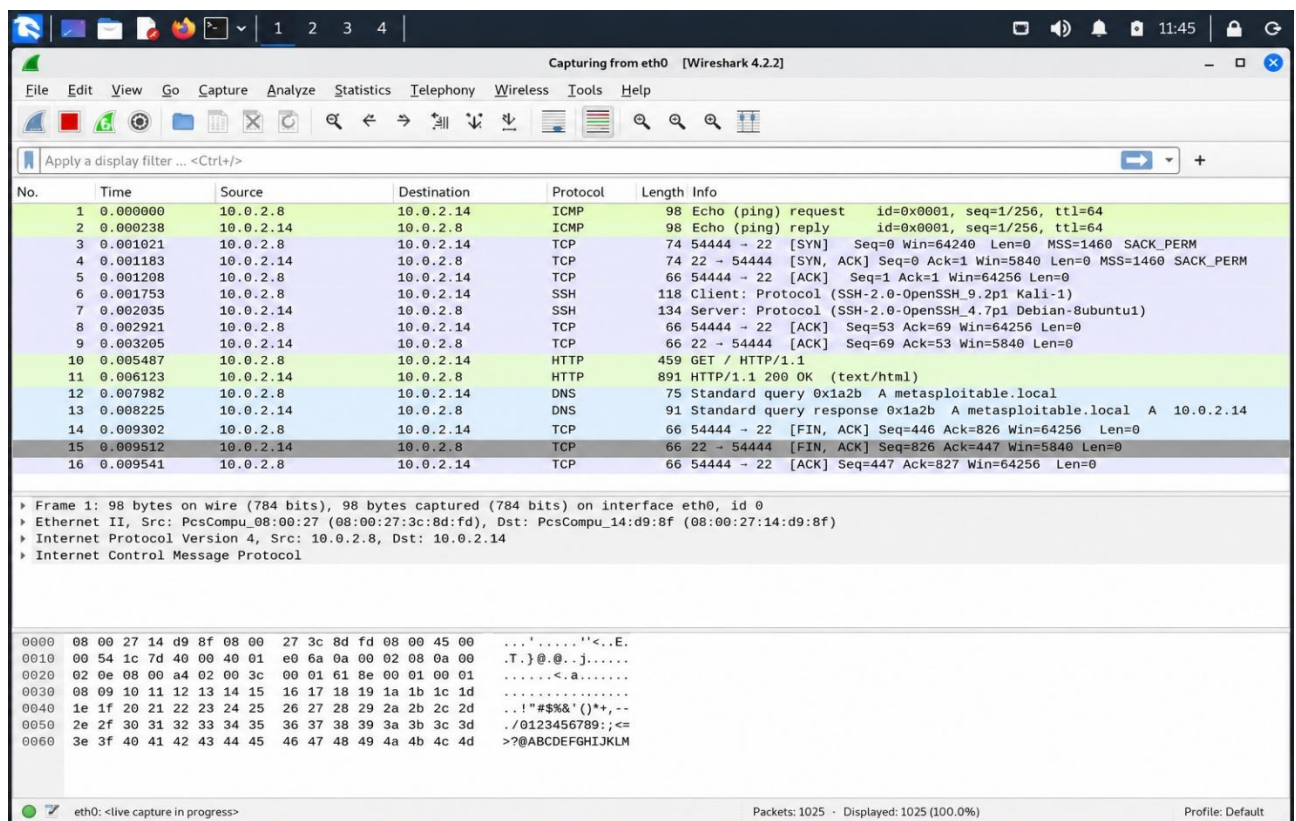
- Exploitation framework
- Vulnerability testing
- Payload delivery and session handling

Wireshark

Wireshark was launched successfully and verified to capture packets on the configured virtual network interface.

Purpose:

- Packet capture and analysis
- Protocol inspection
- Network traffic monitoring



The task involves setting up a controlled penetration testing lab using tools like VirtualBox or VMware to create isolated Kali Linux (attacker) and Metasploitable 2 (target) virtual machines with proper system resources. After installation, the environment is validated by confirming tool functionality such as Nmap, Metasploit Framework, and Wireshark, ensuring they are correctly installed and operational. An initial network reconnaissance is then performed using a basic Nmap scan to identify live hosts and open ports on the target system, establishing a baseline of the attack surface. This baseline is documented and preserved via a VM snapshot so that future security assessments can compare changes, vulnerabilities, and potential exploit paths against the original system state.

2. Observations

- The target machine is reachable within the isolated lab environment.
- Network segmentation successfully isolates testing activities.
- All required security tools are functioning correctly.
- Multiple network services are exposed on the target system.
- The baseline scan establishes the initial attack surface for future vulnerability assessment and exploitation tasks.

3. Conclusion

The penetration testing lab was successfully established using Oracle VirtualBox with Kali Linux as the attacker and Metasploitable 2 as the target. Network connectivity and segmentation were verified through successful communication between the virtual machines. The required security tools, including Nmap, Metasploit Framework, and Wireshark, were validated and confirmed to be operational. An initial baseline assessment identified **23 open TCP ports** and **4 open UDP services**, providing a comprehensive view of the target's attack surface. A virtual machine snapshot was also created to preserve the baseline configuration, ensuring repeatable and controlled testing for future tasks.